
When Shift Happens - Confounding Is to Blame

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Abstract

1 This paper identifies hidden confounding shift as a central mechanism behind several
2 puzzling observations in out-of-distribution (OOD) generalization. Although
3 causal-invariant representations are often viewed as the route to robustness, recent
4 empirical studies show that (i) empirical risk minimization (ERM) can rival or
5 outperform specialized OOD methods and (ii) using all available covariates, rather
6 than only causal covariates, can improve OOD performance. Our first contribution
7 is an explanation: shifts in hidden confounders entangle label, covariate, conditional
8 covariate, and concept shifts, violating assumptions used by many OOD
9 approaches. Our second contribution is a predictive-information analysis proving
10 that, under such shifts, generalization depends on learning environment-specific
11 relationships in addition to invariant structure. Our third contribution is empirical
12 and practical: proxy variables for hidden confounders can improve predictive
13 information and mitigate hidden-confounding-induced failures. Together, these
14 results clarify when ERM and all-covariate models succeed, and they provide
15 reviewer-facing guidance for designing OOD algorithms and covariate-selection
16 strategies under hidden confounding.

1 Introduction

17 A core contribution of this paper is to revisit what makes machine-learning models generalize when
18 deployment environments differ from training environments. Generalization—the ability to draw
19 reliable conclusions about unseen data from observed data—appears across disciplines: in medicine
20 and the social sciences as *external validity* [13], in ecology as *space-for-time substitutions* [60], and
21 in engineering as *robust control* under unmodeled disturbances [35]. Machine learning now supplies
22 many tools for this goal [100, 88, 49], but the paper’s starting point is that the usual robustness story
23 is incomplete when important confounders are hidden.

24 OOD generalization asks a model to maintain performance on in-distribution (ID) data and on test data
25 drawn from different environments. For example, a model trained to predict *food stamp reciprocity*
26 from *household attributes* in one state should remain useful in another state. The literature has
27 developed domain generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81],
28 and robust learning [42, 69, 94] methods that offer guarantees under particular assumptions. The
29 empirical puzzle we foreground is that, under careful model selection [30, 38, 87], standard *empirical*
30 *risk minimization (ERM)* [86] often remains competitive across real applications [30, 24, 50, 57, 67,
31 87]. A second puzzle is that, on 16 tabular datasets, using all available covariates can improve OOD
32 performance even when some covariates are not direct causes of the outcome [57]. Explaining these
33 two observations is one of the paper’s main contributions.

34 Our explanatory contribution centers on a common but often implicit source of distribution shift:
35 hidden confounding. When data come from different environments [71], such as states with different
36 eligibility rules for *food stamp reciprocity*, invariant relationships may appear to be the natural
37 target [6]. Causal invariance is especially attractive because causal relationships are expected to
38 remain stable across environments [71, 59, 6], and many OOD methods seek such structure [6, 59, 31,
39 66, 56, 98, 63]. The limitation we emphasize is that real datasets often omit relevant variables because
40 of data-collection limits, privacy constraints, or measurement error [14, 52, 20, 48]. When omitted
41 variables confound both covariates and outcomes, shifts in their distributions create environment-
42

specific relationships that are not captured by observed causal invariance alone. Ignoring these hidden confounding shifts can degrade OOD performance [40, 4, 85, 62] and can even make models learn the wrong observed covariate–outcome relationship.

Despite this importance, the consequences of hidden confounding shifts for OOD generalization remain underexplored [4, 40, 85, 62]. We therefore make the paper’s contribution structure explicit: the theory explains why hidden confounding changes the objective of generalization, and the experiments show that this explanation is consistent with recent empirical findings [30, 24, 50, 57, 67, 87]. Concretely, the paper makes four contributions that are used throughout the remaining sections:

- **Problem framing.** We isolate hidden confounding shift as a causal mechanism behind multiple observed shifts and use it to explain why informative, non-causal covariates can help OOD prediction (§ 3).
- **Theory.** We decompose predictive information between model predictions and outcomes and show that, under hidden confounding shift, maximizing predictive information requires environment-specific relationships rather than invariant representations alone (§ 4.1, § 4.2).
- **Covariate-selection implication.** We prove that variables informative about the outcome or hidden confounders can improve predictive information, giving a principled reason to consider proxy-like non-causal covariates (§ 4.3).
- **Empirical validation.** On real-world and synthetic datasets, we show evidence that hidden confounding is prevalent, that environment-specific predictive information tracks ID and OOD accuracy, and that informative non-causal covariates improve generalization (§ 5).

Taken together, these contributions explain the surprising effectiveness of ERM under real-world shifts and sharpen the paper’s message for OOD method design: hidden confounding should be modeled directly, not treated as a nuisance left outside the contribution statement.

2 Related work

The related-work contribution of this section is to position hidden confounding shift as a missing explanatory layer across several OOD literatures. OOD generalization includes domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. These areas differ in whether they assume target-domain unlabeled data, environment labels, collaborative training constraints, or detection of unusual samples, but many methods share an emphasis on invariant relationships. Our paper builds on work showing that invariance is not sufficient by itself [46, 72, 93]: in particular, Ye et al. [93] highlight feature *informativeness* as a second requirement. We contribute a hidden-confounding explanation for why both invariance and informativeness matter in OOD prediction.

Proxy variable adjustment: Prior proxy-adjustment work provides the closest technical context for our covariate-selection contribution. When confounders are observed during training but absent at test time, Landeiro and Culotta [40] adjust for confounding shifts to improve classifiers. Building on proxy-based causal-effect adjustment [55, 39], recent domain-adaptation work casts unknown distribution shift as causal-effect identification [4, 85]. OOD generalization under hidden confounding shift has also been studied with overlapping confounder support [62]. Our distinct contribution is explanatory: we show how proxy variables improve OOD generalization through the predictive-information terms that hidden confounding makes salient.

All variable models vs causal models. The all-covariate-versus-causal-covariate debate motivates the empirical side of our contribution. In the benchmark of Nastl and Hardt [57], covariates are grouped as causal, arguably causal, anti-causal, and other spurious variables, and models using all covariates Pareto-dominate causal or arguably causal subsets on ID and OOD data across 16 datasets. Existing theory only partially explains this pattern. Anchor regression [68] balances all-covariate and causal-covariate estimation in linear causal models, while Eastwood et al. [21] show that unstable covariates can help when they carry label information and are conditionally independent of stable covariates given the label. Their method adjusts shift using unlabeled test-domain data, yet ERM remains competitive on a real medical dataset not designed for that setting [8], consistent with Nastl and Hardt [57]. Although MLE can be minimax optimal under well-specified covariate shift [26], our contribution targets the less well-specified real-world case: hidden confounding shift.

3 Manifestations of hidden confounding shift

This section delivers the paper’s first contribution: it makes hidden confounding shift visible as the causal mechanism that can simultaneously generate multiple observed distribution shifts and mislead OOD predictors.

Types of distribution shifts. Table 1 defines the observed shifts that our analysis later measures, but the contribution of this section is to connect them to a single hidden source. Let $\mathbf{X}$ be covariates and Y the target in environments e and e' . If an unobserved variable U causes $\mathbf{X}$, Y , or both, then a change in $\mathbb{P}(U)$ can appear as a change in distributions involving only $\mathbf{X}$ and Y (Figure 1). For example, $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U)\mathbb{P}(\mathbf{X} | U)$, so $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ with $U \rightarrow \mathbf{X}$ implies $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. The same causal accounting applies to shifts in $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$, which is why hidden confounding can unify several shift types that are often studied separately.

The key point for reviewers is that common OOD assumptions correspond to special causal graphs, whereas hidden confounding shift corresponds to the graph in which those assumptions break. If $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), then $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y)\mathbb{P}(\mathbf{X} | Y)$ and changes in $\mathbb{P}(Y)$ create label shift while preserving $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$ because $\mathbb{P}(\mathbf{X} | Y) = \mathbb{P}(\mathbf{X} | Y)$. If $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), then changes in $\mathbb{P}(\mathbf{X})$ create covariate shift while preserving $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. Our setting is the more difficult case in Figure 1 (c), where U causes both $\mathbf{X}$ and Y [50, 4, 40, 85, 62, 65, 64]. Then a shift in $\mathbb{P}(U)$ can make label shift, covariate shift, conditional covariate shift, and concept shift appear together; this is the hidden-confounding configuration summarized in Figure 1. The only stable relation is $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$, known as *stable confounding shift* [85, 4]; because U is hidden, this useful invariance is not directly available to the learner.

Information-theoretic measures: To make this contribution operational, we quantify the shifts above with mutual information following Federici et al. [22]. With environment variable E , the terms $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E|Y)$, $I(Y; E|\mathbf{X})$, $I(\mathbf{X}, Y; E)$ measure shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$ across environments. Label-shift methods often rely on $I(\mathbf{X}; E | Y) = 0$, and covariate-shift methods often rely on $I(Y; E | \mathbf{X}) = 0$. Hidden confounding shift is precisely the case that makes these simplifying zeros unreliable, so these measures become diagnostics rather than assumptions.

Challenges with hidden confounding shift: Figure 2 provides the section’s concrete contribution: a minimal example showing why invariant or pooled prediction can fail and why environment-specific information helps. The outcome Y depends on observed X and hidden confounder U , and environments are induced by shifts in the mean of U , denoted μ_e . A linear regression model trained only on X learns the wrong X – Y relationship, an instance of Simpson’s paradox [73]: within each environment Y increases with X , but the pooled model captures the opposite trend. We observe similar behavior for OOD methods such as IRM [6], VREX [38], and Group DRO [69]. The theory in § 4 explains the remedy: learn environment-specific relationships. The second subplot shows that summary statistics of X by environment, such as mean, standard deviation, and quantiles, help recover the true X – Y relationship, echoing *backdoor adjustment* in causal effect estimation [58]. Because summaries of observed covariates may have limited representational capacity, § 4.3 then studies informative covariates that act as stronger proxies for hidden confounders, as illustrated in the third subplot. The oracle model trained on X and U marks the best-possible mean squared error.

Table 1: Observed distribution shifts used to diagnose the manifestations of hidden confounding shift.

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

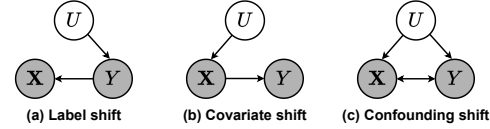


Figure 1: Causal graphs showing when observed shifts can be attributed to label shift, covariate shift, or hidden confounding shift. [89, 83, 25, 5, 47, 96, 18, 79].

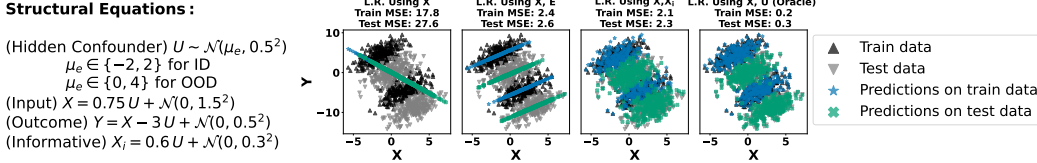


Figure 2: Motivating contribution: hidden confounding shift with minimal environment overlap (distant μ_e) makes four linear regression (L.R.) models behave differently. (i) Training only on X learns the wrong relationship with Y , illustrating Simpson’s paradox. (ii) Adding environment-specific summary statistics of X , denoted E , recovers the correct direction but has limited representation power. (iii) Adding an informative proxy covariate X_i for U improves OOD generalization. (iv) The oracle uses X and U .

4 Impact of hidden confounding shifts on OOD generalization

This section states the paper’s theoretical contribution in the language of predictive information. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ be the prediction for input $\mathbf{X}$, where f is a classifier and ϕ is a feature extractor or transformation. We use entropy $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ and mutual information $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ to make visible which terms OOD objectives encourage or suppress. The risk in environment e is $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X},Y) \sim \mathbb{P}_{\mathbf{X},Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$, and OOD learning seeks a predictor $(f \circ \phi)$ that performs well on ID and OOD data. Existing objectives differ in how they aggregate environment risks: robust optimization [7, 9, 19, 75, 69] minimizes $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$; VREX [38] minimizes $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$; IRM [6] constrains f to be simultaneously optimal across environments while minimizing $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$; and ERM pools environments to minimize $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X},Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$ [6, 38]. Our analysis below explains why these objectives can behave unexpectedly when the missing variable is a hidden confounder.

The theoretical target is predictive information, $I(Y; \hat{Y})$ with $\hat{Y} = (f \circ \phi)(\mathbf{X})$, because it measures how much the prediction tells us about the true label. Reframing OOD generalization this way lets us state the contribution compactly: hidden confounding changes which information terms should be maximized or minimized. We therefore begin with the two representation properties that the decomposition will track—*informativeness* and *invariance* [93].

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informativeness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as $\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

Conditional informativeness is the within-environment information that $\phi(\mathbf{X})$ carries about Y . This term is central to our contribution because hidden confounding makes within-environment relationships predictive even when global invariance is misleading. Minimizing CINF removes useful label information, whereas maximizing it can be undesirable in fairness settings if $\phi(\mathbf{X})$ exploits sensitive or biased environment-specific information. Minimizing $I(\phi(\mathbf{X}); E)$ can reduce such dependence, but it can also reduce predictive performance, as § 4.2 shows. In the setting where E is not treated as sensitive, we analyze when maximizing CINF improves generalization.

Definition 4.2 (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$ across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e., $\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$.*

Invariance measures whether $\phi(\mathbf{X})$ remains consistent across environments for a fixed label Y . This is the term most OOD methods foreground, and our contribution is to show when that foregrounding is incomplete. Preserving too much environment-specific information can overfit, while enforcing invariance can suppress useful confounding-related information. We extend the shift measures from § 3 by using $I(\phi(\mathbf{X}); E)$ for *feature shift* and $I(Y; E | \phi(\mathbf{X}))$ for *concept shift*. Depending on context, *concept shift* denotes either $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$.

4.1 A general decomposition of predictive information

We briefly state the causal-graph vocabulary needed for the decomposition, so the later contribution can be read without ambiguity. A causal graph $\mathcal{G}$ has random variables as nodes and directed

edges as direct causal influences. A *path* between X_i and X_j is a sequence of unique connected nodes, and a *directed path* from X_i to X_j with $i < j$ has all arrows pointing toward X_j , i.e., $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$. Along such a path, X_i is a *parent* of X_{i+1} , X_{i+1} is a *child* of X_i , X_i is an *ancestor* of X_j , and X_j is a *descendant* of X_i . The three basic path structures are a *chain* $X_i \rightarrow X_j \rightarrow X_k$, a *fork* $X_i \leftarrow X_j \rightarrow X_k$, and a *collider* $X_i \rightarrow X_j \leftarrow X_k$. Chains and forks make X_i and X_k marginally dependent but conditionally independent given X_j , i.e. $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k \mid X_j$. Colliders reverse this pattern: $X_i \perp\!\!\!\perp X_k$ marginally, but $X_i \not\perp\!\!\!\perp X_k \mid X_j$ after conditioning on X_j or a descendant. A conditioning set $\mathcal{S}$ *blocks* a path if it contains the middle node of a chain or fork, or if the path contains a collider with neither the collider nor a descendant in $\mathcal{S}$. If all paths from X_i to X_k are blocked by $\mathcal{S}$, then $X_i \perp\!\!\!\perp X_k \mid \mathcal{S}$; otherwise the path is open.

We now express the contribution formally through $I(Y; \hat{Y})$, where $\hat{Y} = (f \circ \phi)(\mathbf{X})$ is based on the learned representation $\phi(\mathbf{X})$. Figure 3 models the relationships among $\mathbf{X}$, Y , U , $\phi(\mathbf{X})$, $\hat{Y}$, and E . The hidden variable U is the confounder, and E indexes shifts in $\mathbb{P}(U)$ across environments. Given either $\mathbf{X}$ or $\phi(\mathbf{X})$, the prediction adds no extra information for reasoning about Y , so $\hat{Y} \perp\!\!\!\perp Y \mid \mathbf{X}$ and $\hat{Y} \perp\!\!\!\perp Y \mid \phi(\mathbf{X})$. We first give a general decomposition that does not specify how U influences $\mathbf{X}, Y$; § 4.2 then specializes the decomposition to hidden confounding.

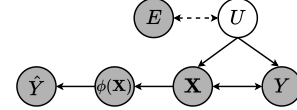


Figure 3: Causal model used for the predictive-information contribution; the bi-directed arrow between $\mathbf{X}$ and Y indicates that some covariates in $\mathbf{X}$ may cause Y , while others may be caused by Y .

Proposition 4.1. For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \frac{\overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}}}{2} + \frac{\overbrace{I(Y; E)}^{\text{Label shift}}}{2} + \frac{\overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}}}{2} - \frac{\overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}}}{2} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

Proofs are in Appendix § A. The decomposition is useful because it maps existing algorithms to the information terms they implicitly target. IRM [6], for example, keeps the classifier f fixed across environments, which relates to maximizing $-I(\phi(\mathbf{X}); E|Y)$: if $\phi(\mathbf{X})$ retains extra environment-specific information conditional on Y , one classifier need not be optimal everywhere. DANN [23] and the independence criterion for fair classification [22] reduce environment-specific information by minimizing $I(\phi(\mathbf{X}); E)$, and domain-adaptation methods likewise learn feature properties invariant to domains [82, 90]. Equation (1) makes the trade-off explicit: minimizing $I(\phi(\mathbf{X}); E)$ may be useful for fairness, but it can reduce $I(Y; \hat{Y})$ and hurt prediction [98, 33, 22]. CDAN [51] targets $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$; this makes E independent of $(\phi(\mathbf{X}), Y)$ and implies $\mathbb{P}(Y \mid \phi(\mathbf{X}), E) = \mathbb{P}(Y \mid \phi(\mathbf{X})) \implies I(Y; E \mid \phi(\mathbf{X})) = 0$. Although CDAN does not directly optimize $I(Y; E \mid \phi(\mathbf{X}))$, Equation (1) reveals why joint-distribution alignment drives this term down. Methods robust to label-distribution shifts address the constant term $I(Y; E)$ [69, 95].

The general decomposition in § 4.1 also clarifies why hidden confounding requires a separate contribution. Equation (1) does not yet specify how U influences $\mathbf{X}$ and Y , and the remaining terms in (1) determine which shift assumptions are plausible. If $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$, then $I(\mathbf{X}; E \mid Y) = 0$, making $I(\phi(\mathbf{X}); E \mid Y)$ a natural quantity to minimize. If $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$, then $I(Y; E \mid \mathbf{X}) = 0$, motivating the minimization of $I(Y; E \mid \phi(\mathbf{X}))$. The hard case—and the one this paper contributes to—is $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$, where the terms in (1) interact rather than vanish independently. The next subsection resolves this case.

4.2 Predictive information under hidden confounding shift

We now state the hidden-confounding contribution directly. When hidden U causes both $\mathbf{X}$ and Y , the predictive-information terms simplify in two important causal cases: (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. For $\mathbf{X} \rightarrow Y$, the inequalities in (2) follow because conditioning on Y opens the collider path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$, allowing additional information flow from $\phi(\mathbf{X})$ to E through Y . By contrast, conditioning on $\phi(\mathbf{X})$ partially blocks the flow from Y to E at $\mathbf{X}$ whenever $\phi(\mathbf{X})$

240 retains information about $\mathbf{X}$.

$$(i) \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

241 For the reverse causal direction $Y \rightarrow \mathbf{X}$, the inequalities in (3) reverse the relevant roles. Conditioning
 242 on Y blocks information flow from $\phi(\mathbf{X})$ to E , while conditioning on $\phi(\mathbf{X})$ opens the collider path
 243 $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ because $\phi(\mathbf{X})$ is a child of $\mathbf{X}$ and conditioning on a collider’s child opens that
 244 path [58] (recall § 4.1).

$$(i) \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

245 Using the inequalities in (2), (3), we obtain the central theoretical contribution: under hidden
 246 confounding shift, the general decomposition collapses to a simpler objective.

247 **Proposition 4.2.** *For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor*
 248 *ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if*
 249 *either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

250 Proposition 4.2 handles $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately, but its implications are especially relevant
 251 for tabular prediction. In recent benchmarks [57, 50, 24], 11 of 16 datasets follow $\mathbf{X} \rightarrow Y$; in
 252 the other 5, covariates caused by Y are fewer than covariates that cause Y . The following boxed
 253 statement makes the proposition’s takeaway explicit.

Contribution takeaway: in the hidden-confounding cases (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$ and
 (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$, the terms *variation, label shift, feature shift, and concept*
shift in (1) cancel as shown in (4). Thus, when either $\mathbf{X} \rightarrow Y$ or $Y \rightarrow \mathbf{X}$, predictive
 information is maximized by maximizing *conditional informativeness* – *residual*, making
 environment-specific predictive content central rather than incidental.

254

255 **Maximizing conditional informativeness:** This term gives a concrete algorithmic reading of
 256 the theory. The MoE approach of Prashant et al. [62] assigns experts to hidden-confounder val-
 257 ues, so each expert improves performance within the environment associated with a confounder
 258 assignment. This is aligned with maximizing $I(\phi(\mathbf{X}); Y|E)$, and Equation (4) supplies a theo-
 259 retical justification for MoE-type behavior under hidden confounding. The same equation also
 260 clarifies the open methodological space: Prashant et al. [62] assume overlapping confounder support,
 261 $\text{supp}(\mathbb{P}(U|\mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U|\mathcal{E}_{tr}))$, and discrete-valued proxies, whereas our framing motivates
 262 methods for broader hidden-confounding settings.

Figure 2 demonstrates the same contribution in a simple low-overlap case: with a continuous-
 valued proxy and no confounding overlap, linear regression can recover the correct covariate-
 target relationship when environment-specific information is supplied. This motivates OOD
 methods that remain effective when confounder support is limited and hidden confounders
 are continuous-valued.

263

264 **Minimizing residual:** The second term in the hidden-confounding objective is the residual
 265 $I(\phi(\mathbf{X}); Y | \hat{Y})$, which measures label information still present in $\phi(\mathbf{X})$ but not captured by $\hat{Y}$. It
 266 can be written as $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The entropy $H(Y | \hat{Y})$ relates
 267 to expected cross-entropy through $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y}) + \text{calibration error}$ [12, 10, 29],
 268 where $H(Y | \hat{Y})$ is the *refinement error*. Thus the contribution suggests two complementary ways to
 269 reduce residual information: directly encourage $Y \perp \phi(\mathbf{X}) | \hat{Y}$, or improve refinement error through
 270 training procedures that prioritize accurate predictions before calibration error [10].

4.3 Impact of additional informative variables

The next contribution turns the theory into a covariate-selection implication. Section 4.2 shows what predictive information requires when the confounder is unobserved; in practice, proxy variables may carry information about that hidden confounder [4, 85, 62]. We therefore analyze how adding non-causal but informative covariates changes the predictive-information terms.

Definition 4.3 (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e., neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$.*

Because $U \rightarrow Y$, a covariate informative about U is also informative about Y , and conversely. The decomposition in (1) shows that improving $I(Y; \hat{Y})$ requires controlling concept shift $I(Y; E \mid \phi(\mathbf{X}))$, and Equation (1) places that term beside the positive conditional-informativeness term. Empirically, Liu et al. [50] suggest that collecting covariates $\mathbf{X}_I$ that make $\mathbb{P}(Y \mid \mathbf{X}, \mathbf{X}_I)$ more stable across environments can improve OOD accuracy. Our theoretical contribution is to specify how informative covariates move the terms in (1): they can increase helpful information while reducing concept shift.

Proposition 4.3. *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned} (i) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y \mid E) &> I(\phi_1(\mathbf{X}); Y \mid E) & (ii) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) &> I(\phi_1(\mathbf{X}); E) \\ (iii) \ I(Y; E \mid \phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) &< I(Y; E \mid \phi_1(\mathbf{X})) & (iv) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) &> I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

Proposition 4.3 makes the covariate-selection contribution explicit. Adding informative covariates increases conditional informativeness (5.i) and feature shift (5.ii), while reducing concept shift (5.iii). The trade-off is that it also increases variation (5.iv), so proxy-rich models may need mechanisms that retain useful environment-specific information while controlling variation.

5 Experimental results

The experiments are organized to validate the paper’s three empirical claims: hidden confounding shift is visible in real benchmarks, the predictive-information terms explain ID and OOD accuracy, and informative covariates improve generalization. We evaluate eight TableShift tabular datasets [24]: *Food stamps, Readmission, Income, Public coverage, Unemployment, Diabetes, Hypertension, and ASSISTments*. Performance is measured with ID and OOD test accuracy. The model set spans two ERM-based methods, XGBoost (XGB) [15] and multi-layer perceptron (MLP); two domain-generalization methods, IRM [6] and VREX [38]; and one robust-learning method, Group DRO (GDRO) [69]. We report averages over 5 random hyperparameter choices as described in Appendix B; random seed effects are statistically insignificant for these datasets [24]. Appendix § B gives the setup details, and Appendix C reports hyperparameter-tuned results.

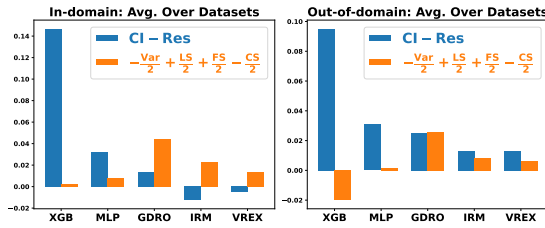
Hidden confounding shift: The first empirical contribution is diagnostic. Hidden confounding shift should manifest through simultaneous changes in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} \mid Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y \mid \mathbf{X})$. Table 2 shows that label shift, covariate shift, conditional covariate shift, and concept shift are all present in the real-world datasets, supporting the hidden-confounding interpretation. As a qualitative complement, we query GPT-4o [1] for plausible hidden confounders in benchmark datasets [24, 57]; Appendix § E reports these analyses. For example, in *food stamps*, unmeasured state-specific *economic policies* may affect both *household income* and *food stamp reciprocity*.

Conditional informativeness vs. accuracy: The second empirical contribution tests the central theoretical quantity from Proposition 4.2. Under hidden confounding shift, predictive information is governed by *conditional informativeness* – *residual*. Figure 4 reports this quantity for the first six datasets in Table 2, which have more than one training environment [57, 24]; Appendix C gives dataset-wise results. The difference *conditional informativeness* – *residual* is positively correlated with ID and OOD accuracy, and XGB has both the strongest performance and the largest value of this difference. The dataset-wise plots also show that the *orange* bars are often close to zero, matching Proposition 4.2: under hidden confounding shift, $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ cancels. The relative ID/OOD ranking is consistent with prior benchmark results [57, 24].

Informative covariates vs. accuracy: The third empirical contribution evaluates the covariate-selection implication. Following [57], we compare three nested covariate sets, *causal covariates* $\subseteq$

Table 2: Empirical diagnostic for hidden confounding shift: all measured shift types are nonzero across real-world datasets. Mean $\pm$ standard deviation is computed over 10 random subsets of 40,000 samples. One-sample t-tests against zero ($\alpha = 0.05$) indicate that all measures are significantly different from zero (p-value ≈ 0).

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: Empirical support for the predictive-information contribution: the difference *conditional informativeness* $-$ *residual* is positively correlated with average ID and OOD test accuracy over the eight datasets summarized in the table.

322 *arguably causal covariates* $\subseteq$ *all covariates* (Appendix § B), and ask how the decomposition terms
323 in (1) change as covariates are added. We summarize this with the *sign consistency metric*: when
324 moving to a larger covariate set, a positive-coefficient term such as conditional informativeness is
325 counted as consistent when it increases with accuracy, while a negative-coefficient term such as
326 concept shift is counted as consistent when it decreases with accuracy. The metric is the fraction
327 of changes aligned with the term’s *beneficial* direction; Appendix § B gives the formal definition.
328 Table 3 shows that conditional informativeness and concept shift both have high sign consistency
329 with accuracy gains. Reducing concept shift matters, but XGB additionally improves conditional
330 informativeness more than the other methods, explaining its extra performance gains. Appendix § C
331 reports dataset-specific results.

332 Because real datasets cannot verify whether *all available covariates* contain *every relevant or*
333 *sufficiently informative* variable, we add a synthetic test where the causal structure is known: $U \rightarrow X$,
334 $U \rightarrow Y$, $X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, with $\mathbf{X}_I$ denoting informative covariates caused by hidden
335 confounder U . This controlled experiment supports the same contribution: adding $\mathbf{X}_I$ improves
336 conditional informativeness and feature shift while reducing concept shift (Figure 5). The variation
337 component stays zero because the problem is simple. Appendix § D gives the data-generation process
338 and additional results.

Table 3: Sign-consistency evidence for the covariate-selection contribution. All methods show high consistency for concept shift (CS), while Proposition 4.2 identifies conditional informativeness (CI) as crucial; XGB is strongest on this term. Res: residual, Var: variation, FS: feature shift, C: causal, AC: arguably causal, A: all.

	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
Method	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

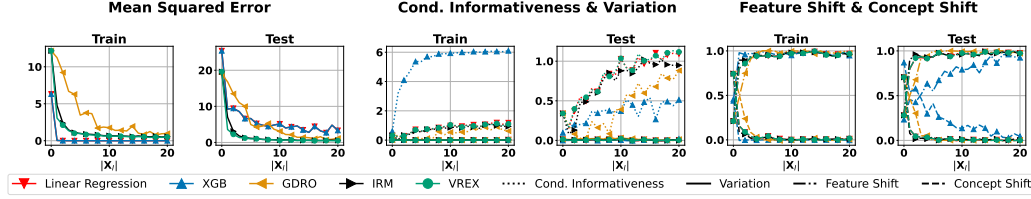


Figure 5: Synthetic covariate-selection result: adding more proxy variables $\mathbf{X}_I$ for U that are informative to Y reduces MSE, increases conditional informativeness and feature shift, and reduces concept shift.

6 Discussion, limitations, and future work

This paper’s contribution is to make hidden confounding shift a first-class explanation for OOD generalization behavior. First, we decompose predictive information between predictions and outcomes, exposing the terms that constrain OOD performance (Equation (1)). Second, we show that hidden confounding entangles covariate, label, concept, and feature shifts (inequalities in (2), (3)), so treating these shifts independently can miss the dominant mechanism. Third, we prove that under hidden confounding shift, predictive information is governed by environment-specific conditional informativeness minus residual information (Equation (4)), making a single invariant predictor insufficient. Fourth, we show that informative proxy covariates can improve the relevant information terms, while acknowledging that no finite observed covariate set can fully remove ambiguity from unobserved confounders. These points summarize the movement from the general decomposition in Equation (1) to the hidden-confounding decomposition in Equation (4). The resulting agenda is clear: develop algorithms for limited confounder overlap, quantify the cost–accuracy trade-off of acquiring informative covariates, handle entangled shifts without untestable proxy assumptions, and model the ambiguity introduced by unobserved confounders.

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Appendix

A Proofs of propositions

Proposition 4.1. For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \frac{\overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}}}{2} + \frac{\overbrace{I(Y; E)}^{\text{Label shift}}}{2} + \frac{\overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}}}{2} - \frac{\overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}}}{2} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

Proof. This proof supports the main decomposition contribution. In the causal substructure $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$ from Figure 3, $\hat{Y}$ adds no information about Y once either $\phi(\mathbf{X})$ or $\mathbf{X}$ is known. The path between $\hat{Y}$ and Y is blocked by conditioning on $\{\phi(\mathbf{X})\}$ or $\{\mathbf{X}\}$, giving $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. We can therefore expand $I(Y; \hat{Y})$ as follows.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

The complementary expansion is:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

Averaging the two equivalent expansions in Equations (6) and (7) isolates the terms used in the paper's decomposition:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

The same decomposition with $\mathbf{X}$ in place of $\phi(\mathbf{X})$ follows by using $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$ and introducing $\mathbf{X}$ in the expansion. $\square$

Proposition 4.2. For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

Proof. This proof establishes the hidden-confounding contribution by showing that the general decomposition collapses in two causal cases from Figure 3: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

653 **Case 1:** $\mathbf{X} \rightarrow Y$. The first inequality follows because conditioning on Y opens the collider path
 654 $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

655 The second inequality follows from the substructure $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$: conditioning on $\phi(\mathbf{X})$
 656 partially blocks information flow from Y to E through $\mathbf{X}$ when $\phi(\mathbf{X})$ encodes information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

657 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1. Applying
 658 the first inequality gives an upper bound:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

659 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1.
 660 Applying the second inequality gives the matching lower bound:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

661 Thus, in this case, the upper and lower bounds coincide: $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq$
 662 $I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

663 **Case 2:** $Y \rightarrow \mathbf{X}$. Here, conditioning on $\phi(\mathbf{X})$ opens the collider path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between
 664 Y and E because $\phi(\mathbf{X})$ is a child of $\mathbf{X}$ [58]. This gives:

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

665 Conditioning on Y also blocks the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$, reducing information flow
 666 between $\phi(\mathbf{X})$ and E :

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

667 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1. Using
 668 the first inequality gives an upper bound:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

669 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.
 670 Using the second inequality gives the matching lower bound:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

671 Thus, in this case, the upper and lower bounds coincide: $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq$
 672 $I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

673 Because the upper and lower bounds coincide in both causal cases, the hidden-confounding decom-
 674 position follows: $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

675 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 676 $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

677 *Proof.* (i) This part proves the conditional-informativeness gain from adding informative covariates.
678 By the chain rule, $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) + I(\mathbf{X}_I; Y|E, \mathbf{X})$. Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$, we
679 have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$, and therefore $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Because $\phi_1(\cdot), \phi_2(\cdot)$ are
680 invertible and mutual information is invariant to invertible transformations, $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) >$
681 $I(\phi_1(\mathbf{X}); Y|E)$.

682 (ii) This part tracks the feature-shift term in Figure 3. The chain rule gives $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) +$
683 $I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, variables informative about Y are informative about U ; with
684 $E \leftrightarrow U$, this yields $I(\mathbf{X}_I; E|\mathbf{X}) > 0$ and hence $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Invertibility of $\phi_1(\cdot), \phi_2(\cdot)$
685 then gives $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

686 (iii) To prove the concept-shift reduction, express $I(Y; E, \mathbf{X}_I|\mathbf{X})$ in two equivalent ways:

$$I(Y; E, \mathbf{X}_I \mid \mathbf{X}) = I(Y; E \mid \mathbf{X}) + I(Y; \mathbf{X}_I \mid E, \mathbf{X}) = I(Y; \mathbf{X}_I \mid \mathbf{X}) + I(Y; E \mid \mathbf{X}_I, \mathbf{X})$$

687 Equating the two expressions and rearranging gives:

$$I(Y; E \mid \mathbf{X}) - I(Y; E \mid \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I \mid \mathbf{X}) - I(Y; \mathbf{X}_I \mid E, \mathbf{X})$$

688 Because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$ and conditioning reduces mutual information unless it opens collider
689 paths, $I(Y; \mathbf{X}_I \mid \mathbf{X}) > I(Y; \mathbf{X}_I \mid E, \mathbf{X})$; here E is neither a collider nor a descendant of a collider.
690 Therefore $I(Y; E \mid \mathbf{X}) > I(Y; E \mid \mathbf{X}_I, \mathbf{X})$. Invertibility and invariance of mutual information yield
691 $I(Y; E \mid \phi_1(\mathbf{X})) > I(Y; E \mid \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

692 (iv) Finally, the variation increase follows from the chain rule: $I(\mathbf{X}, \mathbf{X}_I; E \mid Y) = I(\mathbf{X}; E \mid Y) +$
693 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$ and $E \leftrightarrow U$, informative covariates satisfy $I(\mathbf{X}_I; E|\mathbf{X}, Y) >$
694 0 , so $I(\mathbf{X}, \mathbf{X}_I; E \mid Y) > I(\mathbf{X}; E \mid Y)$. Invertibility of $\phi_1(\cdot), \phi_2(\cdot)$ gives $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) >$
695 $I(\phi_1(\mathbf{X}); E \mid Y)$. $\square$

696 B Experimental setup

697 This appendix section records the experimental details behind the paper’s empirical contribution.
698 Random-seed effects are statistically insignificant in many of these settings [24], so the main paper
699 reports means and standard deviations over five random hyperparameter values sampled from Table B1
700 for each dataset–method pair. To evaluate the predictive-information decomposition on ID and
701 OOD test data, we estimate mutual-information terms with the nonparametric entropy estimation
702 toolbox [37, 77, 78]. Following [24], ID test data come from training domains and OOD test data
703 come from test domains.

704 For consistency, we use 20,000 random samples
705 from each ID and OOD test split when
706 available; some datasets contain fewer samples.
707 This choice does not drive the information-
708 decomposition results: Figure B1 shows that
709 the terms in Equation (1), computed from XGB
710 representations on *income*, remain relatively stable
711 as sample size varies. For neural models,
712 $\phi(\mathbf{X})$ is the layer before the classification head;
713 for XGBoost, we combine output margins and
714 SHAP values. All experiments were run on a
715 single NVIDIA RTX A6000 GPU.

716 To test the covariate-selection contribution, we use the covariate categories of Nastl and Hardt [57]:
717 *causal*, *arguably causal*, and *all covariates*. Causal covariates have well-supported effects on the
718 target with unlikely reverse causation, but relying only on them can omit relevant parents when
719 causal knowledge is incomplete. Arguably causal covariates have less certain status and satisfy
720 at least one criterion: known causal relevance, plausible but potentially bidirectional influence, or
721 likely but unconfirmed causal effect. These groups approximate true causal parents from available
722 knowledge while allowing for confounding. Some datasets also include *anti-causal* covariates
723 that are likely affected by the target, and the *all covariates* set includes every observed variable
724 regardless of causal status. The sign-consistency metric used in § 5 formalizes whether adding
725 covariates moves each information term in its beneficial direction. For each $(m, \sigma_m) \in M$ where

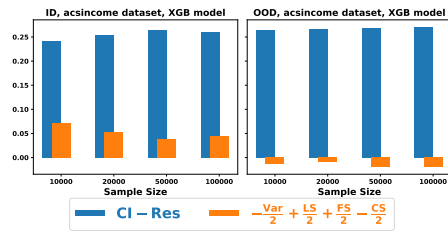


Figure B1: Sample-size check for the empirical contribution: predictive-information terms remain relatively consistent across sample sizes on the *income* dataset.

Table B1: Hyperparameter search space used to support the empirical contribution across ERM, OOD generalization, and robust-learning baselines.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

726 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$
727 $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

728 Here, P contains covariate-set pairs (s, t) such that $t \supset s$; for example, s may be causal covariates
729 and t arguably causal covariates. The factor of 2 accounts for both train and test data. $\mu_s^{\text{tr/te}}(m)$
730 denotes the mean of measure m for covariate set s in training/testing data, and $\mathbb{I}(\cdot)$ is the indicator.

731 With covariate settings $\{causal, arguably\ causal, all\}$, $|P| = 3$. The final sign-consistency score for
732 term m averages over datasets, $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$, where $|D|$ is the number of datasets.

733 C Additional results on real-world datasets

734 This section provides dataset-level support for the main empirical contribution. Figures C2
735 and C3 compare *conditional informativeness – residual* with $-\text{variation} / 2 + \text{label shift} / 2 +$
736 $\text{feature shift} / 2 - \text{concept shift} / 2$ for each dataset; the main paper aggregates these results in Figure 4
737 Left, and this appendix supplies the detailed evidence promised in § C. Consistent with Proposition
738 4.1, the sum of the blue and orange components is positively correlated with model performance.
739 Under potential hidden confounding shift, however, *conditional informativeness – residual*
740 contributes more to predictive information than the combined variation, label-shift, feature-shift, and
741 concept-shift term. These additional results reinforce the main message: conditional informativeness
742 is the salient quantity to improve for OOD generalization under hidden confounding shift.

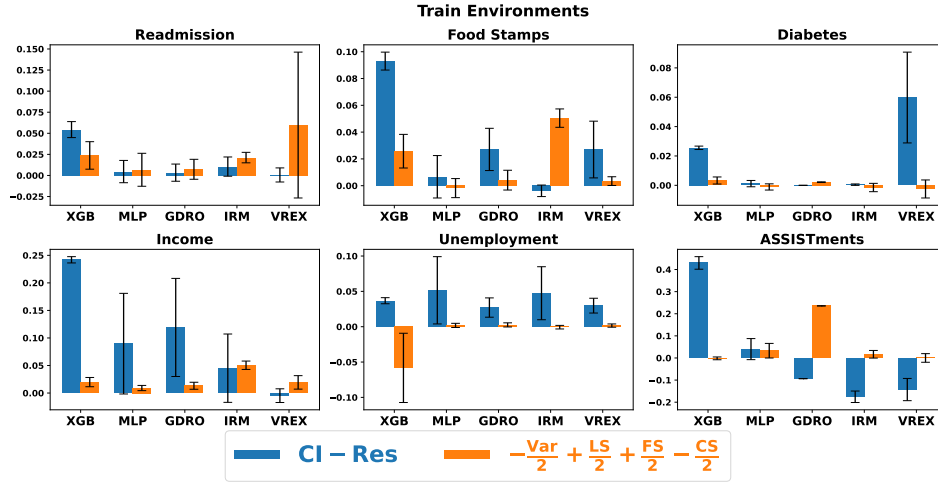


Figure C2: Dataset-specific ID results supporting the predictive-information decomposition.

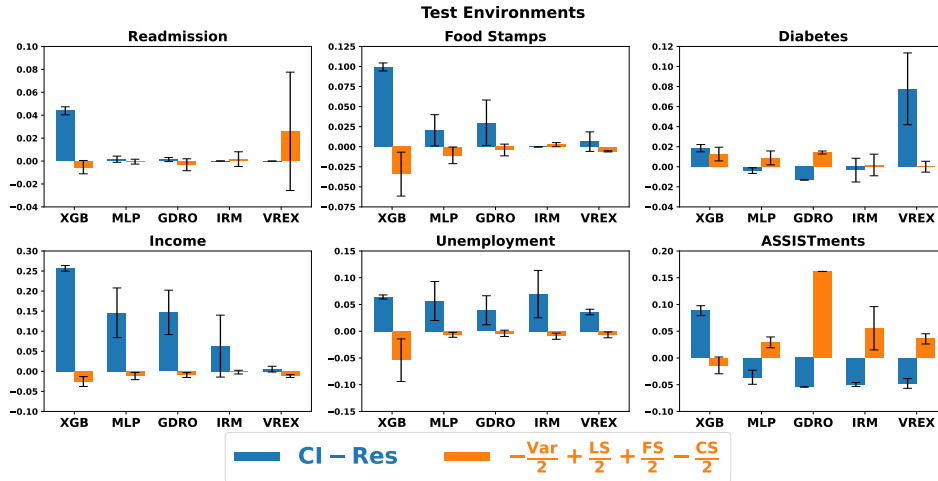


Figure C3: Dataset-specific OOD results supporting the predictive-information decomposition.

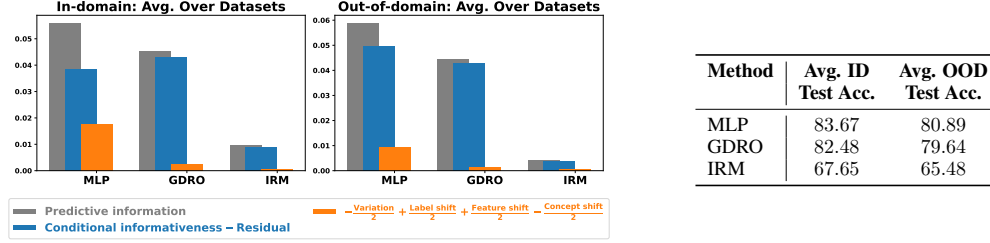


Figure C4: Hyperparameter-tuned results preserve the main contribution: *conditional informativeness - residual* is positively correlated with average ID and OOD test accuracy across the five datasets in the table, and contributes more to predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$.

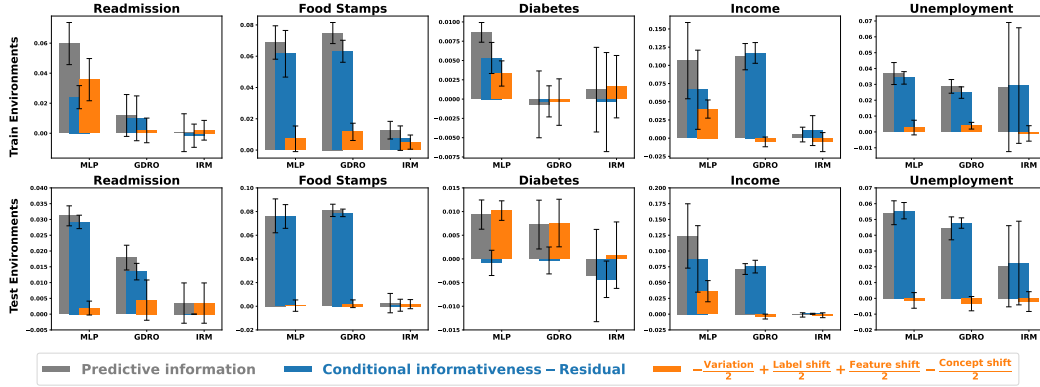


Figure C5: Dataset-specific comparison of predictive-information terms used to trace the contribution across benchmarks.

Table C2: ID and OOD test accuracies used to contextualize the predictive-information comparisons.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 ± 0.14	61.75 ± 0.22	84.74 ± 0.03	82.00 ± 0.06	87.66 ± 0.02	83.22 ± 0.05
GDRO	60.47 ± 1.08	57.60 ± 0.42	84.47 ± 0.04	81.41 ± 0.09	87.47 ± 0.10	82.90 ± 0.10
IRM	50.28 ± 7.35	50.85 ± 1.74	80.91 ± 0.00	78.01 ± 0.00	42.54 ± 36.5	43.48 ± 31.9
	Income		Unemployment			
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.		
MLP	82.92 ± 0.05	81.46 ± 0.01	97.27 ± 0.02	96.04 ± 0.05		
GDRO	82.70 ± 0.05	80.16 ± 0.45	97.28 ± 0.01	96.10 ± 0.00		
IRM	67.91 ± 0.00	60.20 ± 0.00	96.61 ± 0.00	94.84 ± 0.00		

Figures C6- C15 further compare the information-decomposition terms in (1) for each method and dataset. The values in the second column (*variation*) and third column (*feature shift*) are similar in magnitude; based on Equation 1, their difference is therefore close to zero and contributes little to the overall predictive decomposition. Likewise, the fourth column (*label shift*) and fifth column (*concept shift*) are similar in magnitude; based on Equation 1, their difference is also close to zero. This dataset-level pattern supports Proposition 4.2: in almost all settings, the difference *conditional informativeness - residual* is the majority contributor for the predictive information.

Hyperparameter tuning: We also perform hyperparameter tuning on model parameters. We run a grid search over hyperparameter values and select the best hyperparameter values based on ID validation accuracy (not OOD validation accuracy). We run 1500 experiments (5 datasets, 3 models, and 100 hyperparameter choices sampled from a grid of hyperparameter values). In this case, to get the mean and standard deviation results for accuracies and mutual information terms, we run

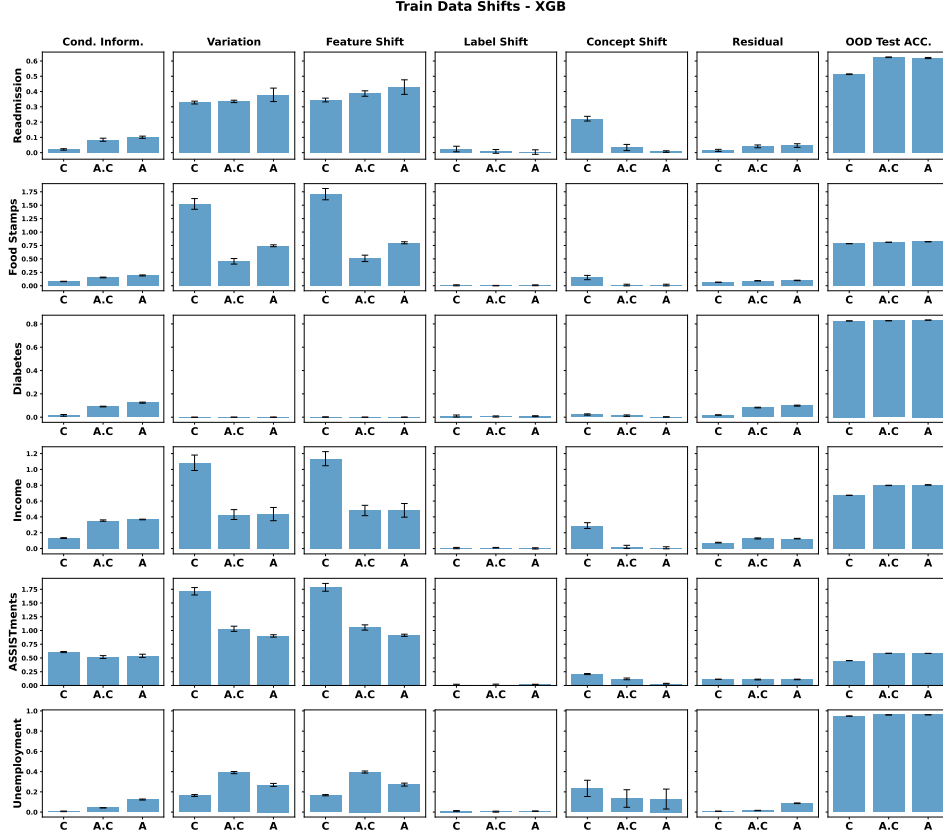


Figure C6: Train-data decomposition for XGBoost across feature groups used to assess the covariate-selection contribution: (C) Causal, (A.C) Arguably causal, (A) All.

each experiment for 5 random seeds, even if the effect of random seeds is insignificant in many cases. Results are shown in Figure C4. From Proposition 4.1, the sum of blue and orange terms is positively correlated with the overall model performance. At the same time, due to potential hidden confounding shift in real-world data, the contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. Dataset-specific results shown in Figure C5 and Table C2 also show similar trends.

Potential limitations with predictive information decomposition analysis: While predictive information $I(Y; \hat{Y})$ measures statistical dependence between predictions and labels, it does not guarantee accuracy. For instance, a binary classifier that systematically predicts the *wrong* label (e.g., $\hat{Y} = 1$ when $Y = 0$) achieves maximal $I(Y; \hat{Y}) = 1$ bit (for binary variables) due to perfect anti-correlation, yet yields 0% accuracy. This occurs because mutual information quantifies *reduction in uncertainty* rather than correctness. Other decomposition terms provide further insight: conditional informativeness $I(\phi(X); Y|E)$ may mask harmful variation $I(\phi(X); E|Y)$, while feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ through spurious correlations. Nevertheless, this decomposition remains valuable for analyzing reasonably performant models, as our experiments demonstrate - it reveals whether predictive information stems from generalizable patterns or environment-specific artifacts, enabling targeted improvements while maintaining interpretability.

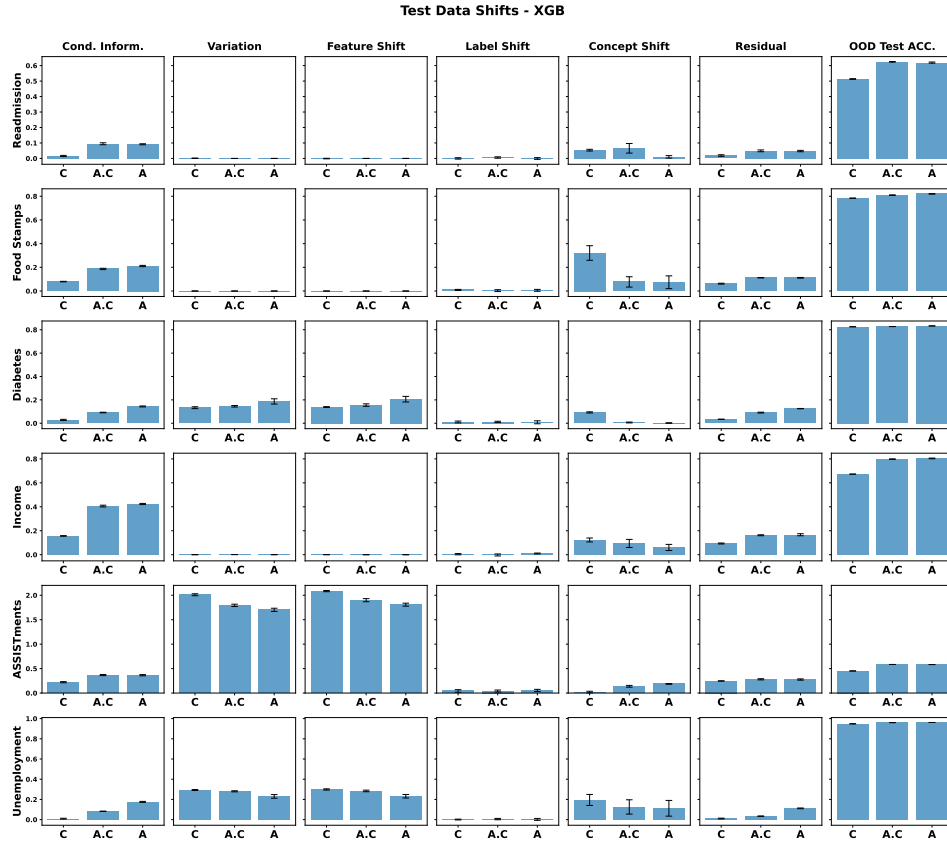


Figure C7: Test-data decomposition for XGBoost across feature groups used to assess the covariate-selection contribution: (C) Causal, (A.C) Arguably causal, (A) All.

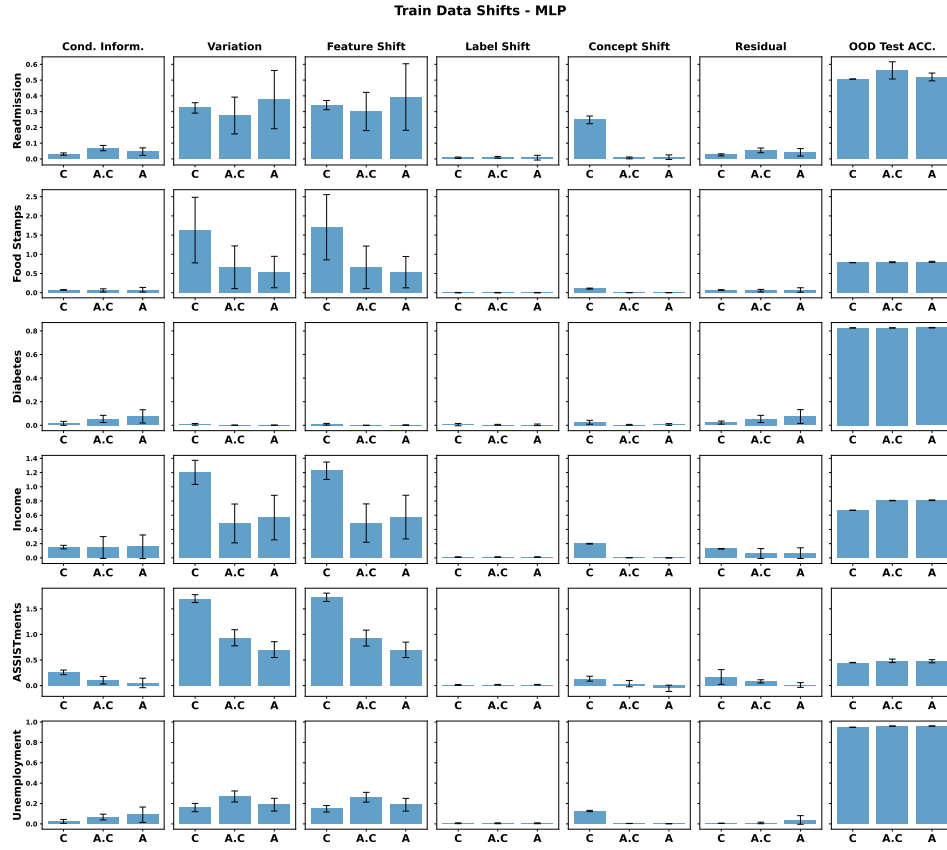


Figure C8: Train-data decomposition for MLP across feature groups used to assess the covariate-selection contribution: (C) Causal, (A.C) Arguably causal, (A) All.

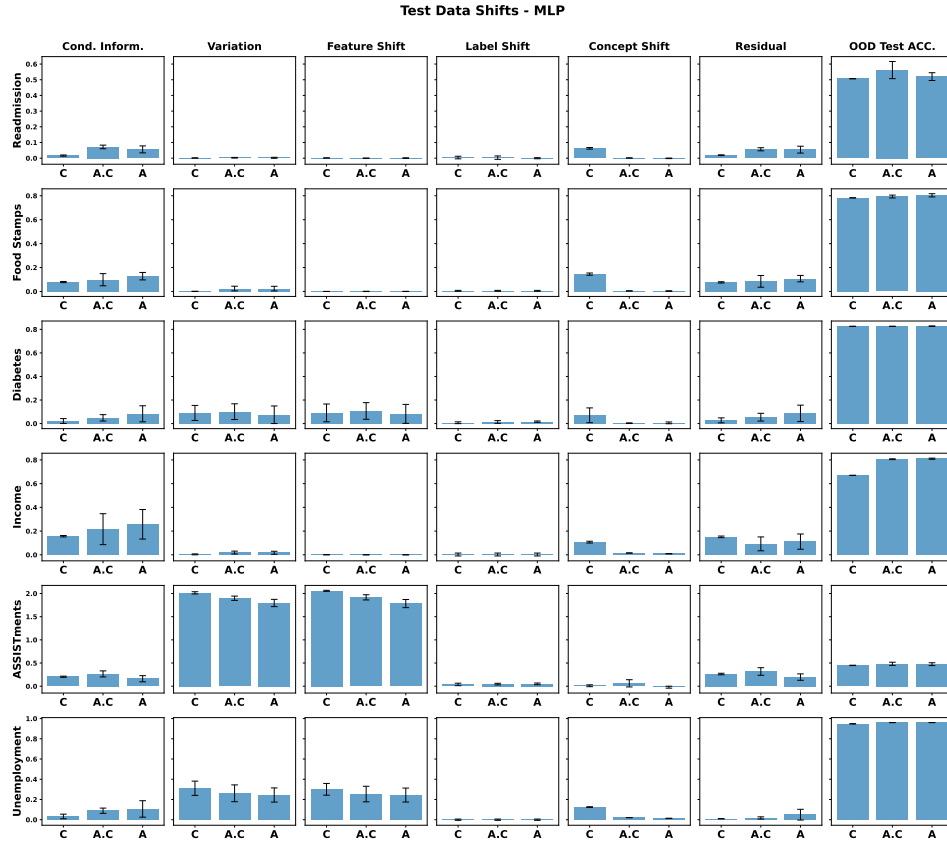


Figure C9: Test-data decomposition for MLP across feature groups used to assess the covariate-selection contribution: (C) Causal, (A.C) Arguably causal, (A) All.

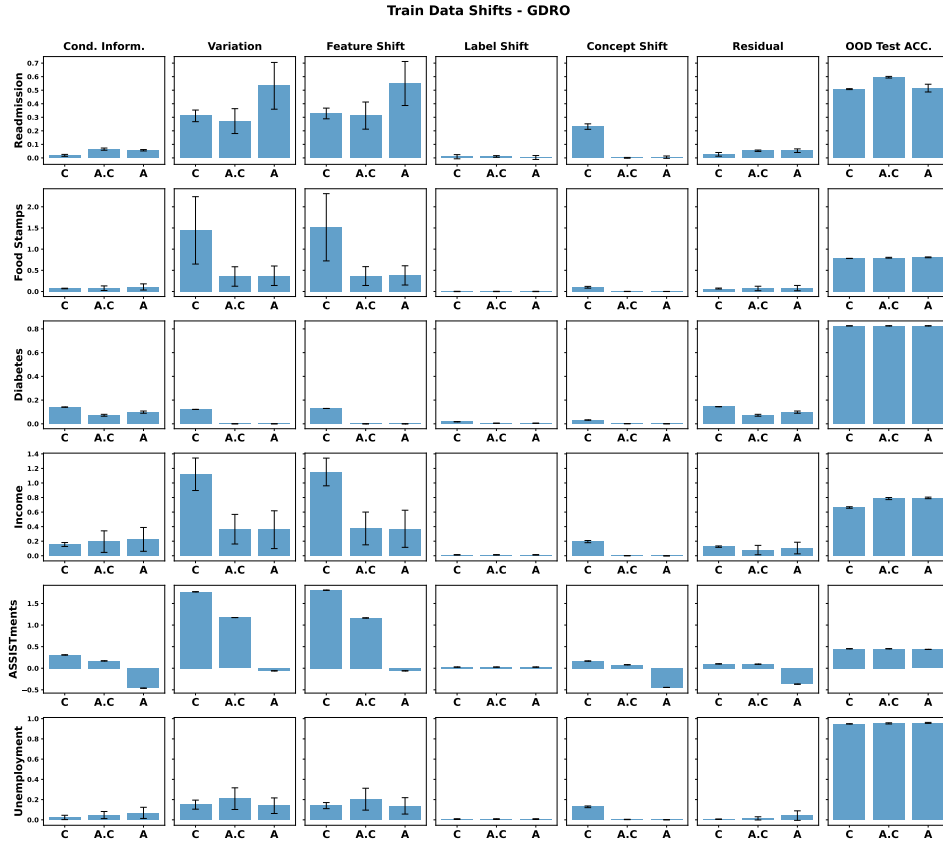


Figure C10: Train-data decomposition for GDRO across feature groups used to assess the covariate-selection contribution: (C) Causal, (A.C) Arguably causal, (A) All.

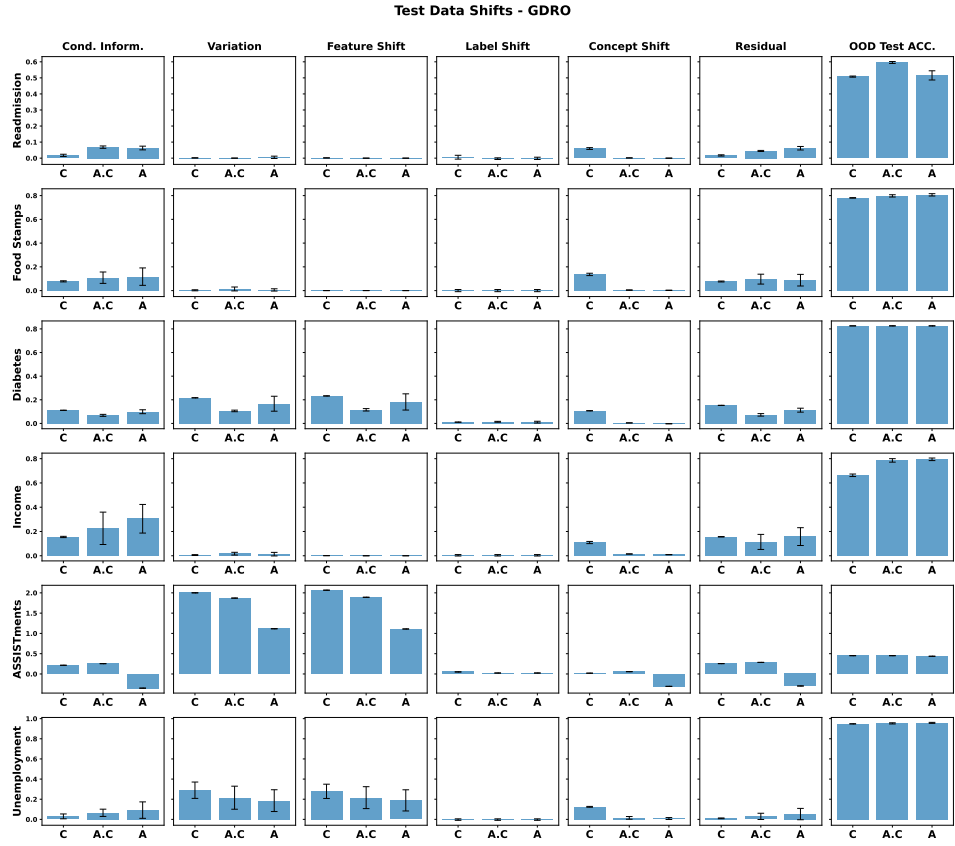


Figure C11: Test-data decomposition for GDRO across feature groups used to assess the covariate-selection contribution: (C) Causal, (A.C) Arguably causal, (A) All.

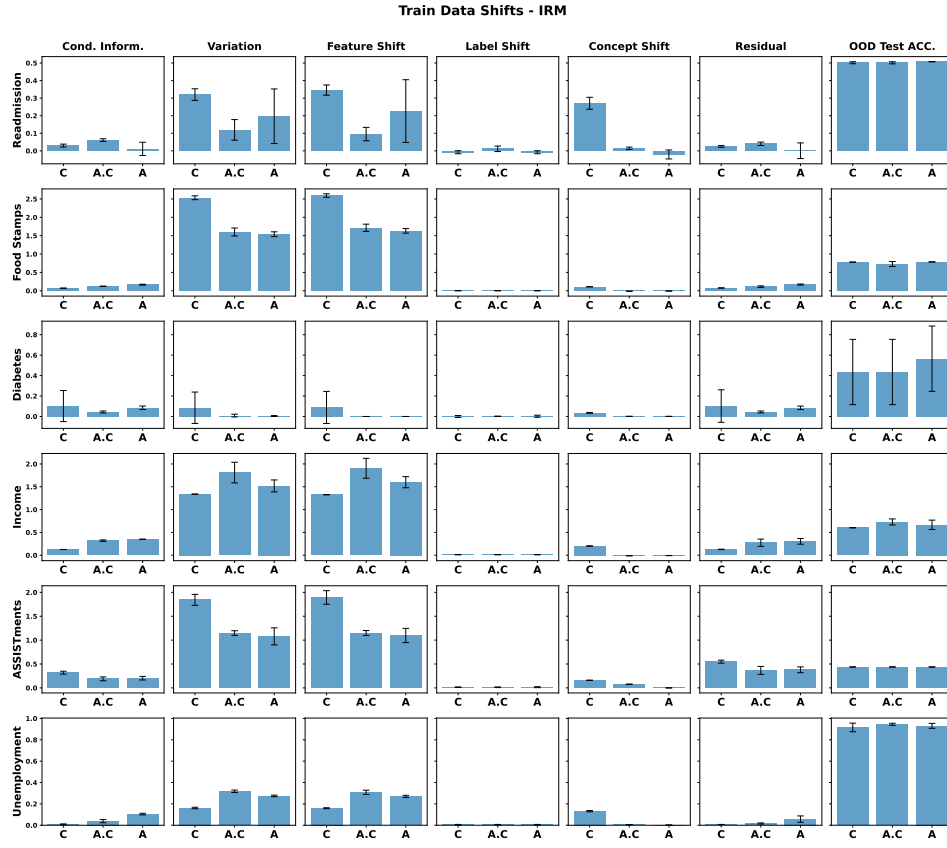


Figure C12: Train-data decomposition for IRM across feature groups used to assess the covariate-selection contribution: (C) Causal, (A.C) Arguably causal, (A) All.

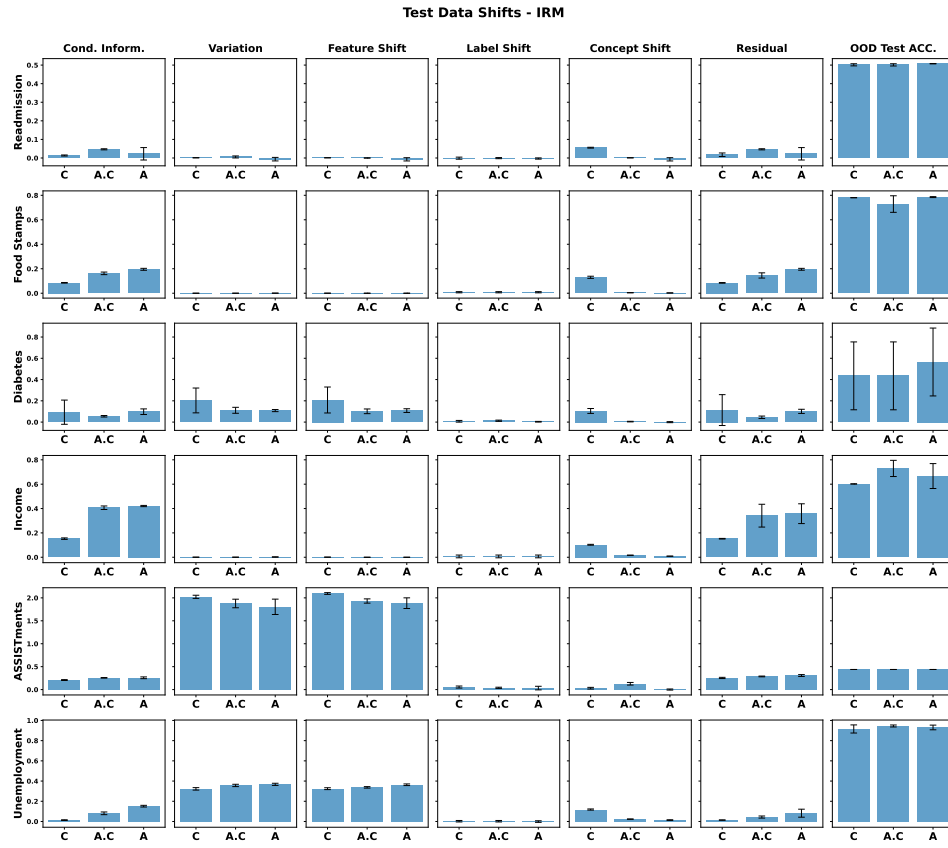


Figure C13: Test-data decomposition for IRM across feature groups used to assess the covariate-selection contribution: (C) Causal, (A.C) Arguably causal, (A) All.

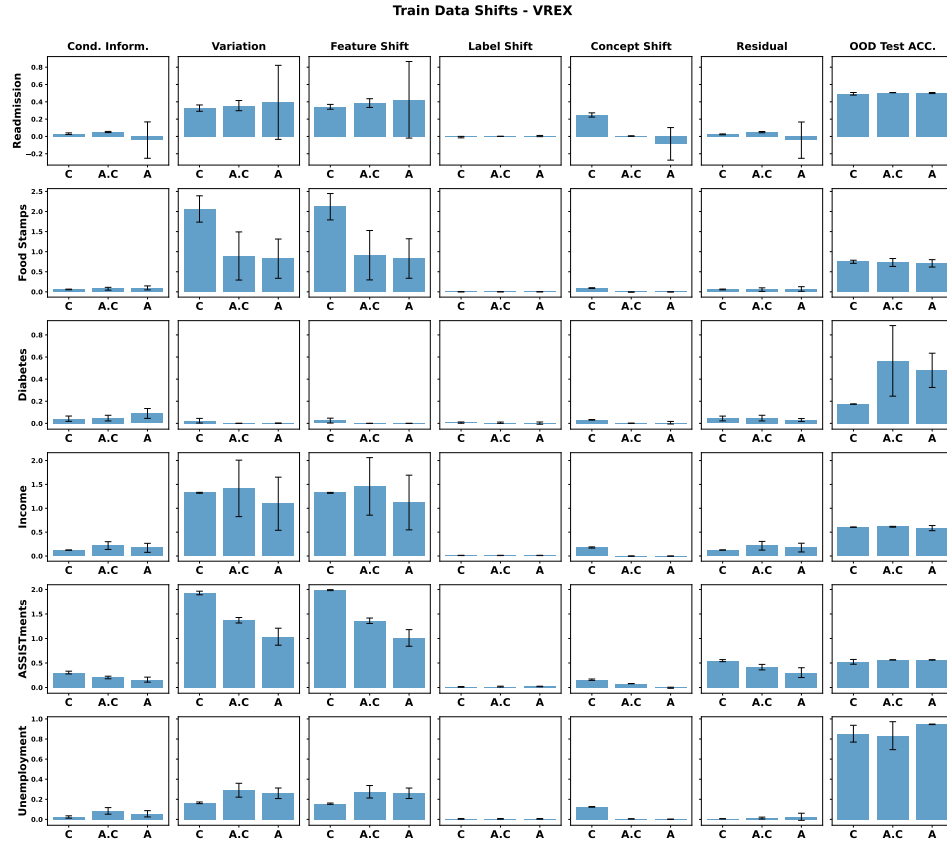


Figure C14: Train-data decomposition for VREX across feature groups used to assess the covariate-selection contribution: (C) Causal, (A.C) Arguably causal, (A) All.

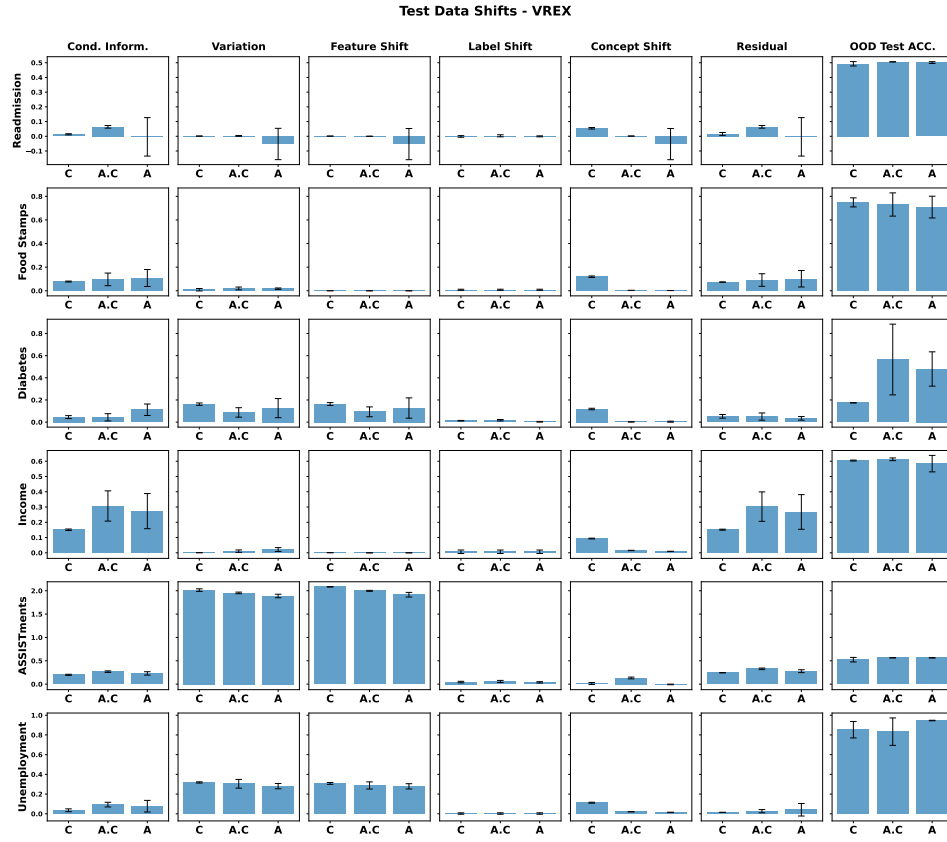


Figure C15: Test-data decomposition for VREX across feature groups used to assess the covariate-selection contribution: (C) Causal, (A.C) Arguably causal, (A) All.

773 D Results on synthetic datasets

774 The synthetic experiments isolate the paper’s contributions in settings where the hidden-confounding
 775 structure is known. We test (i) different informativeness criteria, (ii) the effect of confounder overlap
 776 across environments, and (iii) the effect of varying the number of causal versus anti-causal variables.
 777 The data-generating process follows (12) with causal structure $U \rightarrow X, U \rightarrow Y, U \rightarrow \mathbf{X}_I$, and
 778 $X \rightarrow Y$, where U is unobserved, X is observed, Y is the target, and $\mathbf{X}_I$ are additional informative
 779 covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

780 We vary the functional forms of f_X, f_i , and f_Y to show that the contribution is not tied to a single
 781 functional choice. Domain shifts are created by changing the environment-specific confounder mean
 782 μ_u^e , which lets us control the amount of hidden-confounding shift and evaluate how models respond
 783 across environments.

784 **Informativeness vs. accuracy:** This experiment directly tests the covariate-informativeness con-
 785 tribution. We start with linear structural equations $f_X = 0.3U$, $f_i = 0.1U$, and $f_Y = X - 2U$,
 786 using $\mu_u^e \in \{-2, 2\}$ for training environments and $\mu_u^e \in \{0, 4\}$ for test environments. For Figure 5
 787 in the main paper, one hidden confounder has $|\mathbf{X}_I| = 20$ informative covariates. Increasing the
 788 number of informative covariates from 0 to 20 reduces mean squared error, increases conditional
 789 informativeness, increases feature shift, and decreases concept shift. We then test two additional
 790 informativeness criteria. With one hidden confounder and one informative covariate ($|\mathbf{X}_I| = 1$),
 791 Figure D16 shows that lowering noise σ_i from 2.0 to 0.0 reduces mean squared error, increases con-
 792 ditional informativeness and feature shift, and decreases concept shift. The variation term remains zero
 793 in these synthetic settings. Finally, with 20 hidden confounders and 20 corresponding informative
 794 covariates, using $f_X = 0.3U$, $f_i = 0.2U$, and $f_Y = X - 2U$, Figure D17 shows the same pattern.
 795 Across these variants, informative covariates improve the terms identified by the theory.

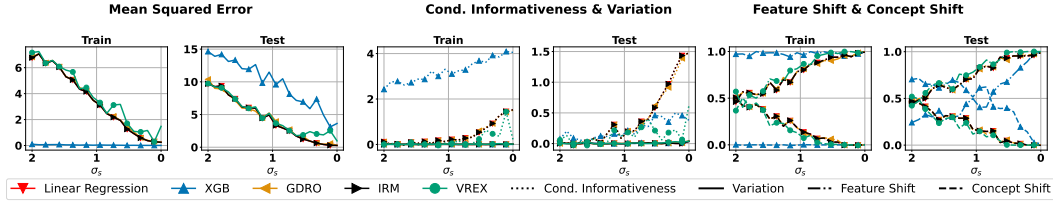


Figure D16: Synthetic informativeness result: a lower-noise proxy variable reduces MSE, increases conditional informativeness and feature shift, and reduces concept shift.

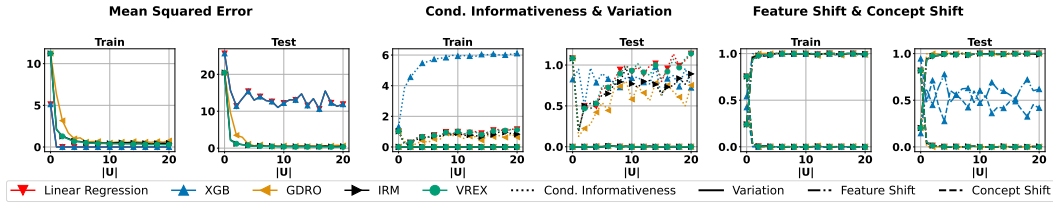


Figure D17: Synthetic multi-confounder result: adding proxy variables $\mathbf{X}_I$ for hidden confounders U that are informative to Y reduces MSE, increases conditional informativeness and feature shift, and reduces concept shift.

796 **Overlapping confounder support vs. accuracy:** This experiment highlights the low-overlap setting
 797 emphasized by the paper’s hidden-confounding framing. Prior OOD work under hidden confounding,
 798 such as Prashant et al. [62], assumes that test confounder support lies within training support. After
 799 showing in linear synthetic data that correct relationships can be learned when observed covariates
 800 contain sufficient confounder information, we extend the test to nonlinear data with MLP and
 801 XGB under low and high confounder-support overlap (Figures D18–D21). In low-overlap settings,

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

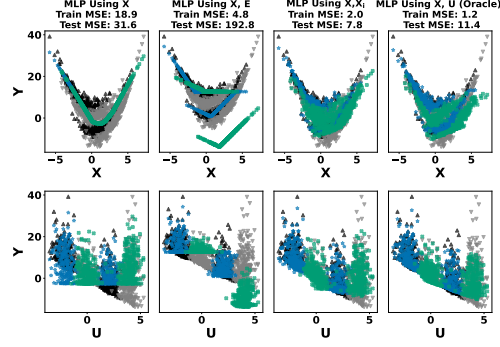


Figure D18: MLP performance in the low-overlap non-linear setting, comparing feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

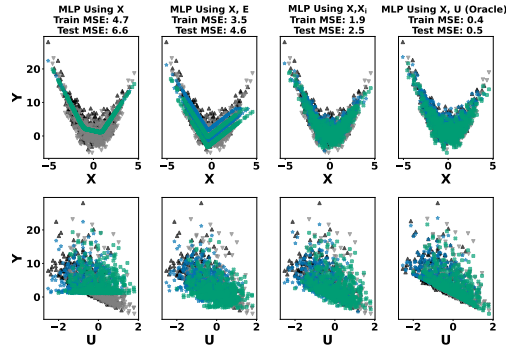


Figure D19: MLP performance in the high-overlap non-linear setting, comparing feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

characterized by distant μ_e between ID and OOD data, models using only X learn an inadequate global function that does not distinguish environments. Environment-specific summary statistics of observed covariates improve performance by encoding environment-specific relationships, but the largest gains come from additional informative covariates. Oracle results and U - Y scatter plots are included to clarify the learned input–output mappings. The expected high-overlap gains appear, but the more salient contribution is that informative covariates also help in the low-overlap regime that challenges common-support assumptions.

Number of causal vs anti-causal variables: This final synthetic study asks whether baseline strength is merely an artifact of tabular tasks often following $X \rightarrow Y$. Recent benchmarks [24] show that baselines such as XGB and MLP often match or outperform OOD methods such as GDRO, IRM,

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

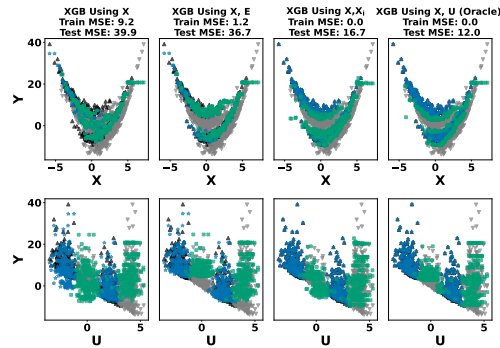


Figure D20: XGB performance in the low-overlap non-linear setting, comparing feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

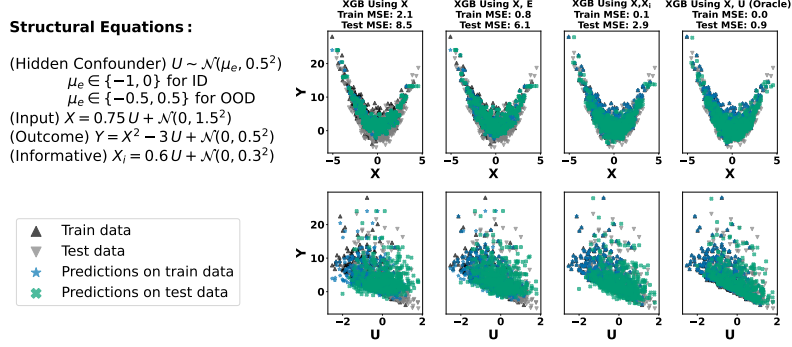


Figure D21: XGB performance in the high-overlap non-linear setting, comparing feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

and VREX, while tabular datasets commonly have the causal structure $\mathbf{X} \rightarrow Y$. We vary the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$; for example, $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$. Figure D22 shows that baselines remain resilient even under $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$, although no method dominates in every setting (Figures D23, D24). Baselines achieve higher PMA-OD scores [24], the fraction of maximum OOD accuracy. We compare ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]), OOD generalization methods (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43], DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]), and robust optimization methods (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

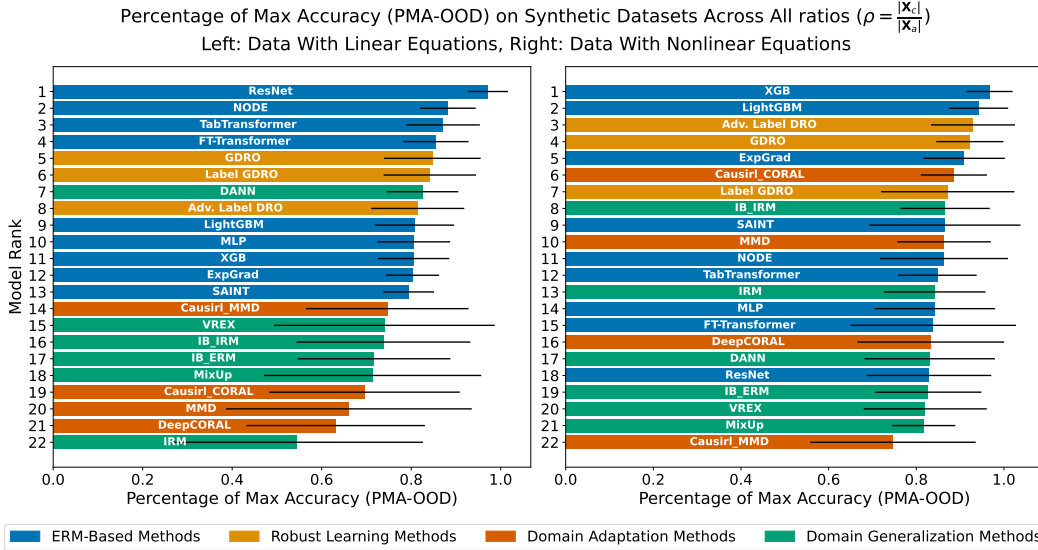


Figure D22: Synthetic PMA-OD result: ERM-based methods achieve a higher percentage of maximum OOD test accuracy, consistent with the real-world findings of [24].

Figure 1 displays the OOD Test Accuracy of various methods across different ρ values (0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0). The methods are categorized into four groups: Domain Generalization Methods (green), Domain Adaptation Methods (orange), Robust Learning Methods (yellow), and ERM-Based Methods (blue). The charts show that ERM-Based Methods generally perform best, followed by Robust Learning Methods, Domain Adaptation Methods, and Domain Generalization Methods. The performance of all methods decreases as ρ increases.

Figure 1 displays the OOD Test Accuracy of various methods across different ρ values (0.0 to 1.0). The methods are categorized into four groups: Domain Generalization Methods (green), Domain Adaptation Methods (orange), Robust Learning Methods (yellow), and ERM-Based Methods (blue). The charts show that Domain Generalization Methods generally perform well at low ρ values, while ERM-Based Methods perform well at high ρ values. Robust Learning Methods show a more consistent performance across different ρ values.

35

E Qualitative analysis of hidden confounding shift in real-world datasets

This qualitative appendix supports the paper’s problem-framing contribution by listing plausible hidden confounders in real-world datasets. Tables E3-E18 identify potential unobserved variables that may influence observed covariates across several datasets [57]; dataset details appear in [24, 57]. We use GPT4o [1] to generate these examples and treat the lists as illustrative rather than exhaustive. The goal is to make the hidden-confounding interpretation concrete across domains, not to claim complete causal discovery.

Prompt: We use the following prompt to query GPT4o for illustrative hidden confounders.

For a target variable, you will be given lists of causal, arguably causal, anti-causal, and spurious covariates. Also, you will be provided with environment variables such that different datasets are collected in environments induced by these environment variables. Provide possible hidden confounders that cause any of the target, causal, arguably causal, spurious, and anti-causal covariates. That is, the hidden confounders should influence the distribution of their children in different domains. Ensure hidden confounders are not a part of the given variables, and your explanation should include how hidden confounders influence their children in different environments.

Table E3: Qualitative hidden-confounding analysis for **Food stamps**: target is food stamp reciprocity in the past year for households with child; environments are U.S. regions.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: Qualitative hidden-confounding analysis for **Income**: target is total person’s income $\geq 56K$ for employed adults; environments are U.S. regions.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: Qualitative hidden-confounding analysis for **Public coverage**: target is coverage of non-Medicare eligible low-income individuals; environments are disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: Qualitative hidden-confounding analysis for **Unemployment**: target is whether a person is unemployed; environments are educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: Qualitative hidden-confounding analysis for **Voting**: target is whether a person voted in the U.S. presidential election; environments are U.S. census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: Qualitative hidden-confounding analysis for **Hypertension**: target is whether a person has hypertension; environments are Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: Qualitative hidden-confounding analysis for **College Scorecard**: target is completion rate for first-time, full-time students at four-year institutions; environments are based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and resources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: Qualitative hidden-confounding analysis for **ASSISTments**: target is whether a student solves a problem correctly on the first attempt in an online learning tool; environments are schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: Qualitative hidden-confounding analysis for **ICU**: target is whether the patient stays in the ICU for longer than 3 days; environments are insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: Qualitative hidden-confounding analysis for **Hospital mortality**: target is whether an ICU patient expires in the hospital during the current visit; environments are insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: Qualitative hidden-confounding analysis for **Childhood lead**: target is blood lead level above the CDC blood level reference value; environments are poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: Qualitative hidden-confounding analysis for **Diabetes**: target is diabetes; environments are preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: Qualitative hidden-confounding analysis for **Sepsis**: target is sepsis onset within the next 6 hours from fine-grained ICU data; environments are lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: Qualitative hidden-confounding analysis for **Hospital readmission**: target is whether a diabetic patient is readmitted within 30 days of initial release; environments are admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: Qualitative hidden-confounding analysis for **MEPS**: target is health care utilization; environments are insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: Qualitative hidden-confounding analysis for **Poverty**: target is household income-to-poverty ratio; environments are citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.